

Python Operators Concept 28 May

- * operator is a symbol which is used to perform some operations on operands
- * operator is deciding the operators result
- * operands are nothing but values or variables which are used in operations

In Python we have several types of operators

- Arithmetic operators
- Relational / comparison
- Logical operators
- Assignment operators
- Short-hand operators
- membership operators
- Identity operators
- ⇒ Bitwise operators

Arithmetic operators

- $+$, $-$, $*$, $/$, $\%$, $**$, $//$ are considered as arithmetic operators
- Arithmetic operators return result in the form of Numbers

→ Arithmetic operators are used to perform some mathematical operations like addition, subtraction,

Ex:-

Relational operators are used to perform some operations

Relational / comparison operators

• $>$, $>=$, $<=$, $=$, $!=$ are considered as a

Relational / comparison operators

- + Relational operators return results in the form of Boolean values like (True / False)
- + Relational operators are used to perform some conditions like equal or not, greater than or not

Logical operators

→ and, or, not keywords are considered as logical operator

• Logical operators return results in the form of Boolean values like True / False

→ Logical operators are used to combine the multiple conditions as a single expression.

Truth table			OR		
Cond 1	Cond 2	Result	T	T	Result
T	T	T	T	F	T
T	F	F	F	T	T
F	T	F	F	F	F
F	F	F			

Order of Priority : $\rightarrow$ not --- and --- or
 first' execut Second third

Exo-
 $10 < 2$ and $30 > 40$ or $5 < 10$ or $30 > 10$ and
 not $20 < 10$

Note :-
 10 and 20 and 30 and $15 \rightarrow 15$ (last Truey value)
 10 and 0 and 30 and $15 \rightarrow 0$ (first Falsey value)
 $()$ and $[]$ and $"$ $\rightarrow ()$ (first Falsey value)

Note
 10 or 20 or 30 or $15 \rightarrow 15$ (first Truey value)
 0 (or) $"$ $"$ 30 or $15 \rightarrow 30$ (first Truey value)
 0 (or) $"$ $"$ or $[]$ or $()$ $\rightarrow$ last Falsey value

Short-hand operators (compound) Augmented

→ Both Arithmetic operators and Assignment Operator combination is nothing but short-hand operator

* $+=$, $-=$, $*=$, $/=$, $\%=$, $**=$, $||=$ are considered as a short-hand operator

$$a = 10$$

$$a = a + 5 = 15$$

$$b = 10$$

$$b += 5 = b = 15$$

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membership operators

→ in, not in are considered as a membership operators

→ By using membership operators we can check given specific value is available (or) not in a specific object

→ in operator Returns True if it is available in given memory otherwise False

→ not in operator returns True if Given specific value is not available in a Given memory otherwise returns True

By using

Member ship operators we can give

Identity

operators

→ is, is not

Identity operators

* is, is not are considered as a Identity operator

→ By using identity operators we can identify the two variable reference are holding the object memory or not